



WOLLO University

Department of software engineering

Exit multiple choice questions for

Data structure and algorithm

Prepared by:

@Neyatibeb

KOMBOLCHA, Ethiopia

2015 E.C

1.

How is an array initialized in C language?

`int a[3] = {1, 2, 3};`

`int a = {1, 2, 3};`

`int a[] = new int[3]`

`int a(3) = [1, 2, 3];`

Hide

Wrong Answer

An array is initialized in C as per the syntax in Option (A).

2.

Which of the following is a linear data structure?

Array

AVL Trees

Binary Trees

Graphs

Hide

Wrong Answer

Array is a linear data structure.

3.

How is the 2nd element in an array accessed based on pointer notation?

`*a + 2`

`*(a + 2)`

`*(a + 2)`

`&(a + 2)`

Hide

Wrong Answer

`a[2]` is equivalent to `*(a + 2)` in pointer notation.

4.

Which of the following is not the type of queue?

Priority queue

Single-ended queue

Circular queue

Ordinary queue

Hide

Wrong Answer

Answer - B) Single Ended Queue is not a type of queue.

5.

From following which is not the operation of data structure?

Operations that manipulate data in some way

Operations that perform a computation

Operations that check for syntax errors

Operations that monitor an object for the occurrence of a controlling event

Hide

Wrong Answer

Answer - A) Operations that manipulate data in some way

6.

What will be the output of the following code snippet?

```
void solve() {  
    int a[] = {1, 2, 3, 4, 5};  
    int sum = 0;  
    for(int i = 0; i < 5; i++) {
```

```
        if(i % 2 == 0) {
            sum += a[i];
        }
    }
    cout << sum << endl;
}
```

5

15

9

6

Hide

Wrong Answer

The code snippet basically calculates the sum of elements at even indices of the array, so we get $1 + 3 + 5 = 9$ as the result.

7.

What will the output of the following code snippet?

```
void solve() {
    int a[] = {1, 2, 3, 4, 5};
    int sum = 0;
    for(int i = 0; i < 5; i++) {
        if(i % 2 == 0) {
            sum += *(a + i);
        }
        else {
            sum -= *(a + i);
        }
    }
    cout << sum << endl;
}
```

2

15

Syntax Error

3

Hide

Wrong Answer

This is an example of accessing an array element through pointer notation. So $*(a + i)$ is equivalent to $a[i]$, and we are adding all the even indices elements, and subtracting all the elements at odd indices, which gives a result of 3.

8.

What is the disadvantage of array data structure?

The amount of memory to be allocated should be known beforehand.

Elements of an array can be accessed in constant time.

Elements are stored in contiguous memory blocks.

Multiple other data structures can be implemented using arrays.

Hide

Wrong Answer

The disadvantage of arrays is that the amount of memory to be allocated should be known beforehand.

9.

How are String represented in memory in C?

An array of characters.

The object of some class.

Same as other primitive data types.

LinkedList of characters.

Hide

Wrong Answer

Strings are represented in C as array of characters.

10.

What is the output of the following code snippet?

```
void solve() {  
    stack<int> s;  
    s.push(1);  
    s.push(2);  
    s.push(3);  
    for(int i = 1; i <= 3; i++) {  
        cout << s.top() << " ";  
        s.pop();  
    }  
}
```

3 2 1

1 2 3

3

1

Hide

Wrong Answer

Stack follows a last in first out methodology, so the elements are popped out in reverse order.

11.

Which of the following is the advantage of the array data structure?

Elements of mixed data types can be stored.

Easier to access the elements in an array

Index of the first element starts from 1.

Elements of an array cannot be sorted

Hide

Wrong Answer

Answer - B) Elements in an array are stored in a contiguous block of memory. So, easier to access the elements in an array

12.

What function is used to append a character at the back of a string in C++?

push_back()

append()

push()

insert()

Hide

Wrong Answer

push_back() function is used for the purpose.

13.

Which one of the following is an application of queue data structure

When a resource is shared among multiple consumers.

When data is transferred asynchronously

Load Balancing

All of the above

Hide

Wrong Answer

Answer - D) All the above statements are the applications of queue data structure.

14.

When a pop() operation is called on an empty queue, what is the condition called?

Overflow

Underflow

Syntax Error

Garbage Value

Hide

Wrong Answer

pop() on an empty queue causes Underflow.

15.

What is the time complexity of the following code snippet in C++?

```
void solve() {  
    string s = "scaler";  
    int n = s.size();  
    for(int i = 0; i < n; i++) {  
        s = s + s[i];  
    }  
    cout << s << endl;  
}
```

O(n)

O(n²)

O(1)

O(log n)

Hide

Wrong Answer

The `s = s + s[i]` line first makes a copy of the original string and then appends the new character in it, leading to each operation being O(n). So the total time complexity is O(n²).

16.

Which of the following data structures can be used to implement queues?

Stack

Arrays

LinkedList

All of the Above

Hide

Wrong Answer

Stack, Arrays, and LinkedList can be used to implement Queues so all the options are correct.

17.

Which of the following data structures finds its use in recursion?

Stack

Arrays

LinkedList

Queues

Hide

Wrong Answer

Stacks find their use in recursion.

18.

Which of the following data structures allow insertion and deletion from both ends?

Stack

Deque

Queue

Strings

Hide

Wrong Answer

Deque is a data structure allowing insertion and deletion from both ends.

19.

What will be the output of the following code snippet?

```
void solve() {
    deque<int> dq;
    for(int i = 1; i <= 5; i++) {
        if(i % 2 == 0) {
            dq.push_back(i);
        }
        else {
            dq.push_front(i);
        }
    }
    for(auto x: dq) {
        cout << x << " ";
    }
    cout << endl;
}
```

1 2 3 4 5

5 4 3 2 1

1 3 5 2 4

5 3 1 2 4

Hide

Wrong Answer

The push_back() function emplaces an element at the back of the deque, and the push_front() function emplaces an element at the front of the deque. In the code snippet, we are alternatively pushing at the front and back of the deque, before printing its contents.

20.

Which of the following sorting algorithms provide the best time complexity in the worst-case scenario?

Merge Sort

Quick Sort

Bubble Sort

Selection Sort

Hide

Wrong Answer

Merge Sort will always have a time complexity of $O(n * \log n)$ which is the best in the worst case among these algorithms.

21.

What is the maximum number of swaps that can be performed in the Selection Sort algorithm?

n - 1

n

1

n - 2

Hide

Wrong Answer

n - 1 swaps are performed at max to sort any array by Selection Sort.

22.

Which of the following is a Divide and Conquer algorithm?

Bubble Sort

Selection Sort

Heap Sort

Merge Sort

Hide

Wrong Answer

Merge Sort is a Divide and Conquer algorithm.

23.

What will be the best sorting algorithm, given that the array elements are small ($\leq 1e6$)?

Bubble Sort

Merge Sort

Counting Sort

Heap Sort

Hide

Wrong Answer

Counting sort sorts an array in $O(n)$ time complexity, taking up an extra space complexity of $O(\max(a[i]))$.

24.

Which of the following are applications of Topological Sort of a graph?

Sentence Ordering.

Course Scheduling.

OS Deadlock Detection.

All of the above.

Hide

Wrong Answer

All the above options are applicable.

25.

Which of the following is known to be not an NP-Hard Problem?

Vertex Cover Problem.

0/1 Knapsack Problem.

Maximal Independent Set Problem.

Travelling Salesman Problem.

Hide

Wrong Answer

The 0/1 Knapsack is not an NP-Hard problem.

26.

Which of the following algorithms are used for string and pattern matching problems?

Z Algorithm

Rabin Karp Algorithm

KMP Algorithm

All of the above

Hide

Wrong Answer

All the above algorithms are used for string and pattern matching.

27.

Consider we have a function, `getLCA()`, which returns us the Lowest Common Ancestor between 2 nodes of a tree. Using this `getLCA()` function, how can we calculate the distance between 2 nodes, given that distance from the root, to each node is calculated?

$\text{dist}(u) + \text{dist}(v) - 2 * \text{dist}(\text{getLCA}(u, v))$

$\text{dist}(u) + \text{dist}(v) + 2 * \text{dist}(\text{getLCA}(u, v))$

$\text{dist}(u) + \text{dist}(v)$

$\text{dist}(u) + \text{dist}(v) - \text{dist}(\text{getLCA}(u, v))$

Hide

Wrong Answer

The distance between 2 nodes, can be broken down into the sum of distances from the root, to each of the nodes. Observe, that in each of these paths, the path from the root to the LCA comes in each of them. So, this needs to be removed from the answer, and hence we get our formula in Option (A).

28.

Which of the following algorithms are useful for processing queries on trees?

Centroid Decomposition.

Heavy Light Decomposition.

Both (A) and (B).

Neither (A) nor (B).

Hide

Wrong Answer

Both (A) and (B) are used to handle queries on trees.

29.

What will the output of the following code snippet be?

```
void solve() {
    vector<int> a = {1, 2, 3, 4, 5};
    sort(a.begin(), a.end(), [&](const int &x, const int &y) {
        return x % 2 < y % 2;
    });
    for(int x: a) {
        cout << x << " ";
    }
    cout << endl;
}
```

1 2 3 4 5

5 4 3 2 1

1 3 5 2 4

2 4 1 3 5

Hide

Wrong Answer

The above code snippet sorts the array, based on a custom comparator. The comparator sorts the array based on the rule, the elements with even parities, have higher priority than the elements with odd parities.

30.

Consider the following code snippet:

```
void solve(vector<int> &a) {
    int queries;
    cin >> queries;
    while(queries--) {
        int type;
        cin >> type;
        if(type == 1) {
            int index, value;
            cin >> index >> value;
            update(a, index, value);
        }
        else {
            int l, r;
            cin >> l >> r;
            cout << getXOR(a, l, r) << endl;
        }
    }
}
```

```
}  
}  
}
```

The update() function updates the element at the given index in the array to some given value. The getXOR() function returns the XOR of the elements in the array a, in the range [l, r]. Which of the following data structures can perform the above tasks optimally?

Segment Trees.

Prefix XOR Arrays.

Tries.

Stacks.

Hide

Wrong Answer

Segment trees would be an optimal choice for such operations.

31.

What is the time complexity of the binary search algorithm?

O(n)

O(1)

O(log2n)

O(n^2)

Hide

Wrong Answer

Time complexity of binary search algorithm is O(log2n).

32.

Kruskal's Algorithm for finding the Minimum Spanning Tree of a graph is a kind of a?

DP Problem.

Greedy Algorithm.

Adhoc Problem.

None of the above.

Hide

Wrong Answer

Kruskal's Algorithm works on the greedy algorithm of taking the lowest weight edges in the MST of a graph unless it forms a cycle.

33.

What will be the output of the following code snippet?

```
void solve() {  
    string s = "00000001111111";  
    int l = 0, r = s.size() - 1, ans = -1;  
    while(l <= r) {  
        int mid = (l + r) / 2;  
        if(s[mid] == '1') {  
            ans = mid;  
            r = mid - 1;  
        }  
        else {  
            l = mid + 1;  
        }  
    }  
    cout << ans << endl;  
}
```

6

7

0

1

Hide

Wrong Answer

This code snippet shows one of the many applications of binary search. Here, we are binary searching for the index of the first occurrence of the character '1' in the given string. When we get the character '1' at the mid index, we store it as the answer and move to the left half which will have the first index of '1' if it occurs. Else we move to the right half. So, the answer will be 7 (0-based indexing).

34.

Maps in C++ are implemented using which of the following data structures?

Red-Black Trees.

Binary Search Trees.

AVL Trees.

Hash Tables.

Hide

Wrong Answer

Red-Black Trees are used to implement Maps in C++.

35.

What will be the output of the following code snippet?

```
void solve() {
    int n = 24;
    int l = 0, r = 100, ans = n;
    while(l <= r) {
        int mid = (l + r) / 2;
        if(mid * mid <= n) {
            ans = mid;
            l = mid + 1;
        }
        else {
            r = mid - 1;
        }
    }
    cout << ans << endl;
}
```

5

4

6

3

Hide

Wrong Answer

The code snippet basically uses binary search to calculate the floor of the square root of a number. Since the square root is an increasing function, so binary search is applicable here. Here, for $n = 24$, the answer is 4.

36.

What is the time complexity of the Sieve of Eratosthenes to check if a number is prime?

$O(n \log(\log n))$ Precomputation, $O(1)$ for check.

$O(n)$ Precomputation, $O(1)$ for the check.

$O(n * \log n)$ Precomputation, $O(\log n)$ for check.

$O(n)$ Precomputation, $O(\log n)$ for check.

Hide

Wrong Answer

The Sieve of Eratosthenes checks if a number is prime in $O(1)$ in the range $[1, n]$ by using an $O(n \log(\log n))$ precomputation and $O(n)$ space complexity.

37.

What will be the output of the following code snippet?

```
int search(int l, int r, int target, vector<int> &a) {
    int mid = (l + r) / 2;
    if(a[mid] == target) {
        return mid;
    }
    else if(a[mid] < target) {
        return search(mid + 1, r, target, a);
    }
    else {
        return search(0, mid - 1, target, a);
    }
}

void solve() {
    vector<int> a = {1, 2, 3, 4, 5};
    cout << search(0, 4, 4, a) << endl;
}
```

3

4

0

2

Hide

Wrong Answer

The above code is just a recursive implementation of binary search which searches for the index of value 4, which is 3.

38.

What is the best case time complexity of the binary search algorithm?

$O(1)$

$O(n)$

$O(\log 2n)$

$O(n^2)$

Hide

Wrong Answer

The best-case time complexity occurs when the target element occurs exactly at the middle of the array and the algorithm terminates in 1 move.

39.

What is the time complexity to insert an element to the front of a LinkedList(head pointer given)?

$O(n)$

$O(1)$

$O(\log n)$

$O(n * \log n)$

Hide

Wrong Answer

We set the next node to the head of the list, and then return that node as the new head.

40.

What is the time complexity to insert an element to the rear of a LinkedList(head pointer given)?

$O(n)$

$O(1)$

$O(\log n)$

$O(n * \log n)$

Hide

Wrong Answer

We need to traverse to the end of the LinkedList and set it next to the new element. So, the traversal will take $O(n)$ time complexity.

41.

What will be the value of “sum” after the following code snippet terminates?

```
void solve(ListNode* root) {  
    /*  
    The LinkedList is defined as:  
    root-> val = value of the node  
    root-> next = address of next element from the node  
    The List is 1 -> 2 -> 3 -> 4 -> 5  
    */  
    int sum = 0;  
    while(root -> next != NULL) {  
        sum += root -> val;  
        root = root -> next;  
    }  
    cout << sum << endl;  
}
```

10

20

5

1

Hide

Wrong Answer

The code calculates the sum of values of the nodes of the given LinkedList, which is 10.

42.

Which of the following can be done with LinkedList?

Implementation of Stacks and Queues

Implementation of Binary Trees

Implementation of Data Structures that can simulate Dynamic Arrays

All of the above

Hide

Wrong Answer

All the above operations can be performed with LinkedList.

43.

What is the information, which a LinkedList's Node must store?

The address of the next node if it exists

The value of the current node

Both (A) and (B)

None of the above

Hide

Wrong Answer

The address of the next node and value of the current node are needed to be stored in a LinkedList's node.

44.

What is the maximum number of children a node can have in an n-ary tree?

2

0

1

n

Hide

Wrong Answer

An n-ary tree can have at most n children.

45.

Worst case time complexity to access an element in a BST can be?

O(n)

O(n * logn)

O(1)

O(logn)

Hide

Wrong Answer

In the worst case, we might need to visit all the nodes in the BST.

46.

Which of the following represents the Postorder Traversal of a Binary Tree?

Left -> Right -> Root

Left -> Root -> Right

Right -> Left -> Root

Right -> Root -> Left

Hide

Wrong Answer

Postorder traversal follows the order of left->right->root in a binary tree.

47.

In what time complexity can we find the diameter of a binary tree optimally?

O(V + E)

O(V)

O(E)

O(V * logE)

Hide

Wrong Answer

We can compute the diameter of a binary tree using a single DFS, which takes the time of O(V + E).

48.

Which of the following statements is true about AVL Trees?

The difference between the heights of left and right nodes cannot be more than 1.

The height of an AVL Tree always remains of the order of O(logn)

AVL Trees are a type of self-balancing Binary Search Trees.

All of the above.

Hide

Wrong Answer

All the above options are applicable for an AVL Tree.

49.

What does the following code snippet calculate (edges represent the adjacency list representation of a graph)?

```
void solve(vector<vector<int>> edges) {
```

```
int count = 0;
for(auto x: edges) {
    for(auto y: x) {
        count += 1;
    }
}
cout << count / 2 << endl;
}
```

Calculates the number of edges in an undirected graph.

Calculates the number of nodes in a given graph.

Calculates the sum of degrees of all nodes in a given graph.

None of the above.

Hide

Wrong Answer

The above code snippet calculates the number of edges in an undirected graph.

50.

In a graph of n nodes and n edges, how many cycles will be present?

Exactly 1

At most 1

At most 2

Depends on the graph

Hide

Wrong Answer

A tree contains by definition n nodes and $n - 1$ edges, and it is an acyclic graph. When we add 1 edge to the tree, we can form exactly one cycle by adding this edge.

51.

A node in a tree, such that removing it splits the tree into forests, with size of each connected component being not greater than $n / 2$ is called?

Center

Diameter

Centroid

Path

Hide

Wrong Answer

Such a node in a tree is called a centroid of the tree.

52.

What does the following code snippet do?

```
void dfs(int node, vector<vector<int>> &edges, vector<bool> &vis,
vector<int> &dp) {
    vis[node] = true;
    for(auto x: edges[node]) {
        if(!vis[x]) {
            dp[x] = dp[node] + 1;
            dfs(x, edges, vis, dp);
        }
    }
}
```

Stores depths of all the nodes in a given tree, with respect to some root node.

Counts the number of nodes in a given tree.

Finds the diameter of a tree.

Checks if all the nodes are reachable in a given tree.

Hide

Wrong Answer

The above code snippets finds the depth of each node, in the tree, with respect to some root node, and stores them in the dp array, using a dfs traversal on the tree.

53.

Which of the following algorithms are used to find the shortest path from a source node to all other nodes in a weighted graph?

BFS.

Dijkstra's Algorithm.

Prims Algorithm.

Kruskal's Algorithm.

Hide

Wrong Answer

The Dijkstra's algorithm is used to find the shortest path from a source node to all other nodes in a weighted graph.

54.

What is the best time complexity we can achieve to precompute all-pairs shortest paths in a weighted graph?

$O(n^3)$

$O(n^2)$

$O(n)$

$O(n^4)$

Hide

Wrong Answer

The Floyd-Warshall Algorithm computes All Pairs Shortest Paths in a weighted graph, in $O(n^3)$.

55.

Which data structure is mainly used for implementing the recursive algorithm?

Queue

Stack

Array

List

Hide

Wrong Answer

Answer - B) Stacks are used for the implementation of recursion.

Q 1 - In order traversal of binary search tree will produce –

A - unsorted list

B - reverse of input

C - sorted list

D - none of the above

Hide Answer

Q 2 - Postfix expression is just a reverse of prefix expression.

A - True

B - False

Answer : B

Explanation

Expression notations are not reverse (or so) of each other, rather operators used in the expression have different arrangements.

[Hide Answer](#)

Q 3 - The following formula will produce

$$F_n = F_{n-1} + F_{n-2}$$

A - Armstrong Number

B - Fibonacci Series

C - Euler Number

D - Prime Number

Answer : B

Explanation

Fibonacci Series generates subsequent number by adding two previous numbers.

[Hide Answer](#)

Q 4 - Minimum number of spanning tree in a connected graph is

A - n

B - n^{n-1}

C - 1

D - 0

Answer : C

Explanation

Every connected graph at least has one spanning tree.

[Hide Answer](#)

Q 5 - Which method can find if two vertices x & y have path between them?

A - Depth First Search

B - Breadth First Search

C - Both A & B

D - None A or B

Answer : C

Explanation

By using both BFS and DFS, a path between two vertices of a connected graph can be determined.

[Hide Answer](#)

Q 6 - Which of the below mentioned sorting algorithms are not stable?

A - Selection Sort

B - Bubble Sort

C - Merge Sort

D - Insertion Sort

Answer : A

Explanation

Except selection sort, all other sorting algorithms are stable.

[Hide Answer](#)

Q 7 - The Θ notation in asymptotic evaluation represents –

A - Base case

B - Average case

C - Worst case

D - NULL case

Answer : A

Explanation

Θ represents average case. O represents worst case and Ω represents base case.

[Show Answer](#)

Q 8 - All possible spanning trees of graph G

A - have same number of edges and vertices.

B - have same number of edges and but not vertices.

C - have same number of vertices but not edges.

D - depends upon algorithm being used.

Answer : A

Explanation

All possible spanning trees of graph G, have same number of edges and vertices.

Show Answer

Q 9 - A pivot element to partition unsorted list is used in

A - Merge Sort

B - Quick Sort

C - Insertion Sort

D - Selection Sort

Answer : B

Explanation

The quick sort partitions an array using pivot element and then calls itself recursively twice to sort the resulting two subarray.

Hide Answer

Q 10 - An adaptive sorting algorithm –

A - adapts to new computers.

B - takes advantage of already sorted elements.

C - takes input which is already sorted.

D - none of the above.

Answer : B

Explanation

A sorting algorithm is said to be adaptive, if it takes advantage of already 'sorted' elements in the list that is to be sorted.

1. The 'C' language is developed by ---

A) Ken martin

- B) Thomson
- C) Charles Babbage
- D) Dennis Ritchie

Ans: D

2. There is the --- number of keywords in 'C' language.

- A) 30
- B) 32
- C) 31
- D) 35

Ans: B

3. --- is any well-defined computational procedure that takes a value or a set of values as input and produces an output.

- A) Algorithm
- B) Flowchart
- C) Program
- D) process

Ans: A

4. The algorithm should be effective which means it must have ---

- A) basic primitive
- B) basic steps
- C) complex primitive
- D) complex steps

Ans: A

5. Which of these statements is used for multi-way branching?

- A) switch
- B) for
- C) if
- D) auto

Ans: A

6. Which statement is used to end the loop and move the control to the next statement outside the loop?

- A) continue
- B) break
- C) exit
- D) switch

Ans: B

7. Array is also known as ---

- A) keyword
- B) reserved word
- C) subscripted variable
- D) simple variable

Ans: C

8. --- is the process of re-arranging a given set of objects in a specific order.

- A) searching

- B) instantiation
- C) sorting
- D) swapping

Ans: C

9. Every 'C' program can be thought of as a collection of ---

- A) Keywords
- B) functions.
- C) variables
- D) constants

Ans: B

10. To use the library function `atan2()` which header file must be included in the program?

- A) `math.h`
- B) `string.h`
- C) `process.h`
- D) `conio.h`

Ans: A

11. Pointer arithmetic is possible for `++` and `--`, but it has no meaning of `++` and `--`.

- A) multiplication, division, addition, subtraction
- B) addition, subtraction, multiplication, division
- C) preincrement, predecrement, postincrement, postdecrement
- D) postincrement, postdecrement, preincrement, predecrement

Ans: B

12. `__` is a special type of variable which is used to hold the address of the variable.

- A) structure
- B) union
- C) pointer
- D) void

Ans: C

13. Which of the following is invalid file open mode?

- A) `wb+`
- B) `ab+`
- C) `rb+`
- D) `wr+`

Ans: D

14. Which of these functions allows you to write individual characters to a stream?

- A) `fputc()`
- B) `putc()`
- C) both A) and B)
- D) none of these

Ans: C

15. --- is a representation of data and the operations allowed on that data.

- A) program
- B) data structure
- C) data type
- D) structure

Ans: B

16. A data structure is said to be --- if its elements form a sequence.

- A) bipolar
- B) linear
- C) polar
- D) scalar

Ans: B

17. A stack is very useful in situations when data have to be stored and then retrieved in the --- order.

- A) same
- B) reverse
- C) ascending
- D) descending

Ans: B

18. --- is useful in the evaluation of the arithmetic expression.

- A) stack
- B) queue
- C) tree
- D) Graph

Ans: A

19. which of these principles is used in a stack?

- A) Last in first out
- B) First in first out
- C) Last in last out
- D) None of these

Ans: A

20. A stack is defined formally as a list in which all insertion and deletion are made at ---

- A) same time
- B) same end
- C) different end
- D) both A) and B)

Ans: B

21. Linked lists permit insertion and removal of nodes at any point in the list in constant time, but do not allow ---.

- A) random access.
- B) displaying items
- C) searching

D) all of these

Ans: A

22. --- is a self-referential data type because it contains a pointer or link to another datum of the same type.

A) Stack

B) Queue

C) A linked list

D) None of these

Ans: C

23. In a singly linked list the link field of --- contains a null pointer.

A) first node

B) any node

C) middle node

D) the last node

Ans: D

24. In --- the link field of the last node contains a null pointer.

A) stack

B) queue

C) singly linked list

D) circular linked list

Ans: C

25. A node that has both a parent and at least one child is called ---

A) Nonleaf

B) leaf

C) root

D) root

Ans: A

26. A strictly binary tree in which the number of nodes at any level I is ---, the tree is said to be a complete binary tree.

A) $2^i - 1$

B) $2^i - 1$

C) $2^i - 1$

D) $2^i + 1$

Ans: B

27. In the case of the AVL tree, the lookup takes --- time.

A) $n \log n$

B) $O(\log n)$

C) $O(n \log n)$

D) $\log n$

Ans: B

28. A tree can also be represented using an array, which is called --- representation

A) asymmetric

B) random

- C) symmetric
- D) sequential

Ans: D

29. When the edges in a graph have no direction, the graph is called--- graph

- A) undirected.
- B) simplex
- C) duplex
- D) nondirected

Ans: A

30. A directed graph is ---if at least two vertices are not connected.

- A) strongly disconnected
- B) weakly connected
- C) weekly disconnected
- D) strongly connected

Ans: B

31. The outdegree of a vertex in a digraph is ---

- A) number of arcs leaving the vertex.
- B) number of arcs entering the vertex
- C) total no of vertex connected to that vertex
- D) total no of vertex disconnected to that vertex

Ans: A

32. If every vertex of any graph has the same degree, then the graph is called ---

- A) directed graph
- B) undirected graph
- C) regular graph.
- D) subgraph

Ans: C

33. Most of the algorithms studied by computer scientists that solve problems are kinds of ---

- A)search algorithms.
- B) sort algorithms
- C) data structures
- D) None of these

Ans: A

34. The set of all possible solutions to a problem is called the ---

- A) Database
- B) data structure
- C) search space
- D) None of these

Ans: C

35. Informed search algorithms use --- functions to apply knowledge about the structure of the search space to try to reduce the amount of time spent in searching.

- A) built-in
- B) heuristic
- C) user-defined
- D) Complex

Ans: B

36. The binary search method needs no; more than --- comparisons.

- A) $\lceil \log_2 n \rceil + 1$
- B) $\log_2 n$
- C) $\lceil \log_2 n \rceil - 1$
- D) $\lceil \log_2 n \rceil + 2$

Ans: A

37. Let $P=50$ after executing the following instruction what will be the value of P and Q.

$Q = p++;$

- A) $P = 51$ $Q = 50$
- B) $P = 51$ $Q = 51$
- C) $P = 50$ $Q = 50$
- D) $P = 50$ $Q = 51$

Ans: A

38. From the following list identify a 'C' keyword.

- i) for
- ii) while
- iii) auto
- iv) shortest
- v) long

- A) i), ii) and v)
- B) i) and v)
- C) i), iii) and v)
- D) i), ii), iv) and v)

Ans: C

39. The pictorial representation of the algorithm is called ---

- A) Dataflow
- B) process flow
- C) Flowchart
- D) process chart

Ans: C

40. The rectangle symbol in the flowchart represents ---

- A) Decision making
- B) start or end of the flowchart
- C) Processing
- D) None of these

Ans: C

41. The goto statement is the --- statement

- A) conditional control

- B) unconditional control
- C) looping
- D) both A) and C)

Ans: B

42. Which is an entry-controlled loop?

- A) do-while
- B) switch
- C) continue
- D) while

Ans: D

43. What is an array?

- A) collection of variables of a different type.
- B) collection of variables of the same type
- C) collection of data objects of the same type
- D) collection of data objects of different type

Ans: C

44. C automatically converts any intermediate value to the proper types so that the expression can be evaluated without losing any significance. This conversion is called---

- A) typecasting
- B) swapping
- C) data conversion
- D) instantiation

Ans: A

45. The function fmod (variable/value, variable/value) returns ---

- A) the remainder of two floating-point value
- B) quotient of two floating-point value
- C) the remainder of two integer value
- D) quotient of two integer value

Ans: A

46. What is the user-defined function?

- A) Function available in the header file
- B) function available in 'C' compiler
- C) a function was written by the user
- D) either A) or B)

Ans: C

47. The process of allocating memory at run time is called ---

- A) static memory allocation
- B) dynamic memory allocation
- C) pre memory allocation
- D) post-memory allocation

Ans: B

48. Which of these functions are used for dynamic memory allocation?

- A) calloc()

- B) realloc
- C) malloc()
- D) all of these

Ans: D

49. Which of these functions is used to read a string from a stream?

- A) fgetc()
- B) fgetws()
- C) getc()
- D) fputc()

Ans: B

50. Identify file open function.

- A) fopen()
- B) freopen()
- C) tmpfile()
- D) all of these

Ans: D

1. The 'C' language is developed by ---

- A) Ken martin
- B) Thomson
- C) Charles Babbage
- D) Dennis Ritchie

Ans: D

2. There is the --- number of keywords in 'C' language.

- A) 30
- B) 32
- C) 31
- D) 35

Ans: B

3. --- is any well-defined computational procedure that takes a value or a set of values as input and produces an output.

- A) Algorithm
- B) Flowchart
- C) Program
- D) process

Ans: A

4. The algorithm should be effective which means it must have ---

- A) basic primitive
- B) basic steps
- C) complex primitive

D) complex steps

Ans: A

5. Which of these statements is used for multi-way branching?

A) switch

B) for

C) if

D) auto

Ans: A

6. Which statement is used to end the loop and move the control to the next statement outside the loop?

A) continue

B) break

C) exit

D) switch

Ans: B

7. Array is also known as ---

A) keyword

B) reserved word

C) subscripted variable

D) simple variable

Ans: C

8. --- is the process of re-arranging a given set of objects in a specific order.

A) searching

B) instantiation

C) sorting

D) swapping

Ans: C

9. Every 'C' program can be thought of as a collection of ---

A) Keywords

B) functions.

C) variables

D) constants

Ans: B

10. To use the library function at of() which header file must be included in the program?

A) math.h

B) string.h

C) process.h

D) conio.h

Ans: A

11. Pointer arithmetic is possible for --- and ---, but it has no meaning of --- and ---.

A) multiplication, division, addition, subtraction

B) addition, subtraction, multiplication, division

- C) preincrement, predecrement, postincrement, postdecrement
- D) postincrement, postdecrement, preincrement, predecrement

Ans: B

12. --- is a special type of variable which is used to hold the address of the variable.

- A) structure
- B) union
- C) pointer
- D) void

Ans: C

13. Which of the following is invalid file open mode?

- A) wb+
- B) ab+
- C) rb+
- D) wr+

Ans: D

14. Which of these functions allows you to write individual characters to a stream?

- A) fputc()
- B) putc()
- C) both A) and B)
- D) none of these

Ans: C

15. --- is a representation of data and the operations allowed on that data.

- A) program
- B) data structure
- C) data type
- D) structure

Ans: B

16. A data structure is said to be --- if its elements form a sequence.

- A) bipolar
- B) linear
- C) polar
- D) scalar

Ans: B

17. A stack is very useful in situations when data have to be stored and then retrieved in the

--- order.

- A) same
- B) reverse
- C) ascending
- D) descending

Ans: B

18. -- is useful in the evaluation of the arithmetic expression.

- A) stack
- B) queue
- C) tree
- D) Graph

Ans: A

19. which of these principles is used in a stack?

- A) Last in first out
- B) First in first out
- C) Last in last out
- D) None of these

Ans: A

20. A stack is defined formally as a list in which all insertion and deletion are made at --

- A) same time
- B) same end
- C) different end
- D) both A) and B)

Ans: B

21. Linked lists permit insertion and removal of nodes at any point in the list in constant time, but do not allow --.

- A) random access.
- B) displaying items
- C) searching
- D) all of these

Ans: A

22. -- is a self-referential data type because it contains a pointer or link to another datum of the same type.

- A) Stack
- B) Queue
- C) A linked list
- D) None of these

Ans: C

23. In a singly linked list the link field of -- contains a null pointer.

- A) first node
- B) any node
- C) middle node
- D) the last node

Ans: D

24. In -- the link field of the last node contains a null pointer.

- A) stack
- B) queue
- C) singly linked list

D) circular linked list

Ans: C

25. A node that has both a parent and at least one child is called ---

A) Nonleaf

B) leaf

C) root

D) root

Ans: A

26. A strictly binary tree in which the number of nodes at any level i is ---, the tree is said to be a complete binary tree.

A) $2^i - 1$

B) 2^{i-1}

C) $2^i - 1$

D) 2^{i+1}

Ans: B

27. In the case of the AVL tree, the lookup takes --- time.

A) $n \log n$

B) $O(\log n)$

C) $O(n \log n)$

D) $\log n$

Ans: B

28. A tree can also be represented using an array, which is called --- representation

A) asymmetric

B) random

C) symmetric

D) sequential

Ans: D

29. When the edges in a graph have no direction, the graph is called--- graph

A) undirected.

B) simplex

C) duplex

D) nondirected

Ans: A

30. A directed graph is ---if at least two vertices are not connected.

A) strongly disconnected

B) weakly connected

C) weekly disconnected

D) strongly connected

Ans: B

31. The outdegree of a vertex in a digraph is ---

A) number of arcs leaving the vertex.

B) number of arcs entering the vertex

C) total no of vertex connected to that vertex

D) total no of vertex disconnected to that vertex

Ans: A

32. If every vertex of any graph has the same degree, then the graph is called ---

- A) directed graph
- B) undirected graph
- C) regular graph.
- D) subgraph

Ans: C

33. Most of the algorithms studied by computer scientists that solve problems are kinds of ---

- A) search algorithms.
- B) sort algorithms
- C) data structures
- D) None of these

Ans: A

34. The set of all possible solutions to a problem is called the ---

- A) Database
- B) data structure
- C) search space
- D) None of these

Ans: C

35. Informed search algorithms use --- functions to apply knowledge about the structure of the search space to try to reduce the amount of time spent in searching.

- A) built-in
- B) heuristic
- C) user-defined
- D) Complex

Ans: B

36. The binary search method needs no; more than --- comparisons.

- A) $\lceil \log_2 n \rceil + 1$
- B) $\log_2 n$
- C) $\lceil \log_2 n \rceil - 1$
- D) $\lceil \log_2 n \rceil + 2$

Ans: A

37. Let $P=50$ after executing the following instruction what will be the value of P and Q.

$Q = p++;$

- A) $P = 51$ $Q = 50$
- B) $P = 51$ $Q = 51$
- C) $P = 50$ $Q = 50$
- D) $P = 50$ $Q = 51$

Ans: A

38. From the following list identify a 'C' keyword.

- i) for
 - ii) while
 - iii) auto
 - iv) shortest
 - v) long
- A) i), ii) and v)
B) i) and v)
C) i), iii) and v)
D) i), ii), iv) and v)

Ans: C

39. The pictorial representation of the algorithm is called --

- A) Dataflow
B) process flow
C) Flowchart
D) process chart

Ans: C

40. The rectangle symbol in the flowchart represents --

- A) Decision making
B) start or end of the flowchart
C) Processing
D) None of these

Ans: C

41. The goto statement is the -- statement

- A) conditional control
B) unconditional control
C) looping
D) both A) and C)

Ans: B

42. Which is an entry-controlled loop?

- A) do-while
B) switch
C) continue
D) while

Ans: D

43. What is an array?

- A) collection of variables of a different type.
B) collection of variables of the same type
C) collection of data objects of the same type
D) collection of data objects of different type

Ans: C

44. C automatically converts any intermediate value to the proper types so that the expression can be evaluated without losing any significance. This conversion is called--

- A) typecasting
- B) swapping
- C) data conversion
- D) instantiation

Ans: A

45. The function fmod (variable/value, variable/value) returns ---

- A) the remainder of two floating-point value
- B) quotient of two floating-point value
- C) the remainder of two integer value
- D) quotient of two integer value

Ans: A

46. What is the user-defined function?

- A) Function available in the header file
- B) function available in 'C' compiler
- C) a function was written by the user
- D) either A) or B)

Ans: C

47. The process of allocating memory at run time is called ---

- A) static memory allocation
- B) dynamic memory allocation
- C) pre memory allocation
- D) post-memory allocation

Ans: B

48. Which of these functions are used for dynamic memory allocation?

- A) calloc()
- B) realloc
- C) malloc()
- D) all of these

Ans: D

49. Which of these functions is used to read a string from a stream?

- A) fgetc()
- B) fgetws()
- C) getc()
- D) fputc()

Ans: B

50. Identify file open function.

- A) fopen()
- B) freopen()
- C) tmpfile()
- D) all of these

Ans: D